Graph-Laplacian Water-Quality Modeling

Seasonality, MERIT Hydro Covariates, Graph Structure, and Leave-One-Node-Out Prediction

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1 Modeling Overview

The objective is to predict water-quality measurements at monitoring locations distributed across a river network.

For river node i and time t, the general model separates the prediction into temporal, chemical, physical, and graph-based components:

$$\boxed{y_{i,t} = X_{i,t}\beta + S_i\gamma + g_i + \varepsilon_{i,t}.}$$

Here,

- $y_{i,t}$ is the observed water-quality value at node i and time t;
- $X_{i,t}$ contains time-varying predictors, including other water-quality variables and seasonal terms;
- S_i contains static physical characteristics of the river location;
- g_i is a graph-based spatial effect;
- $\varepsilon_{i,t}$ is residual error.

The model therefore combines

$$\boxed{\text{chemistry} + \text{seasonality} + \text{physical river characteristics} + \text{river-network structure.}}$$

2 Seasonal Predictors

Water-quality variables can vary substantially with both the time of day and the time of year.

Rather than treating hour or day-of-year as ordinary linear variables, cyclic Fourier predictors are used.

The general Fourier pair for seasonal period m is

$$\boxed{\sin\left(\frac{2\pi kt}{m}\right), \qquad \cos\left(\frac{2\pi kt}{m}\right),}$$

where k is the harmonic number.

2.1 Interpretation of the Harmonic Order K

The value K determines how many sine/cosine harmonic pairs are included.

For example, if

$$K = 1,$$

then only

$$\sin\left(\frac{2\pi t}{m}\right), \qquad \cos\left(\frac{2\pi t}{m}\right)$$

are included.

This produces a relatively simple smooth seasonal cycle.

If

$$K = 2,$$

the model additionally includes

$$\sin\left(\frac{4\pi t}{m}\right), \qquad \cos\left(\frac{4\pi t}{m}\right).$$

For

$$K = 4,$$

the seasonal component becomes

$$\boxed{\sum_{k=1}^{4}\left[a_k \sin\left(\frac{2\pi kt}{m}\right) + b_k \cos\left(\frac{2\pi kt}{m}\right)\right].}$$

Increasing K therefore allows the seasonal curve to have a more complex shape.

A larger K does not simply increase the magnitude of seasonality. Instead, it increases the **shape flexibility** of the seasonal pattern.

2.2 Daily Seasonality

Let h_t denote hour of day.

For daily seasonality,

$$m = 24.$$

The daily Fourier component is

$$\boxed{F_{\text{day}}(t) = \sum_{k=1}^{K_{\text{day}}} \left[a_k \sin\left(\frac{2\pi k h_t}{24}\right) + b_k \cos\left(\frac{2\pi k h_t}{24}\right) \right].}$$

When

$$K_{\text{day}} = 1,$$

the model represents one basic daily cycle.

Higher values allow the daily cycle to depart from a simple sinusoidal shape.

2.3 Annual Seasonality

Let d_t denote day of year.

The annual component is

$$\boxed{F_{\text{year}}(t) = \sum_{k=1}^{K_{\text{year}}} \left[c_k \sin\left(\frac{2\pi k d_t}{365.25}\right) + e_k \cos\left(\frac{2\pi k d_t}{365.25}\right) \right].}$$

The value K_{year} determines how flexible the annual seasonal pattern can be.

2.4 Why Sine and Cosine Terms Are Used

Time-of-day and day-of-year are circular variables.

For example,

$$23:00$$

and

$$00:00$$

are adjacent in time even though their numerical values 23 and 0 appear far apart.

The transformation

$$\left(\cos\left(\frac{2\pi t}{m}\right), \sin\left(\frac{2\pi t}{m}\right)\right)$$

maps time onto a circle, preserving this cyclic structure.

2.5 Daily–Annual Seasonal Interactions

For water temperature, additional improvement was obtained by allowing the daily cycle to change throughout the year.

The interaction component uses products such as

$$\sin\left(\frac{2\pi k h_t}{24}\right)\sin\left(\frac{2\pi d_t}{365.25}\right),$$

$$\sin\left(\frac{2\pi k h_t}{24}\right)\cos\left(\frac{2\pi d_t}{365.25}\right),$$

$$\cos\left(\frac{2\pi k h_t}{24}\right)\sin\left(\frac{2\pi d_t}{365.25}\right),$$

and

$$\cos\left(\frac{2\pi k h_t}{24}\right)\cos\left(\frac{2\pi d_t}{365.25}\right).$$

These terms allow

$$\boxed{\text{daily behavior in winter} \neq \text{daily behavior in summer.}}$$

This is different from simply increasing K because the interaction allows the daily seasonal cycle itself to vary across the annual cycle.

3 MERIT Hydro Physical River Predictors

The river network was constructed using MERIT Hydro raster data.

Several physical characteristics from the MERIT data were attached to river nodes and tested as predictors.

The main variables were

$$\boxed{\text{Elevation,} \quad \text{Upstream Area,} \quad \text{River Width,} \quad \text{HAND,} \quad \text{Local Slope.}}$$

3.1 Elevation

Elevation is represented by

$$z_i = \text{elevation of river node } i.$$

Elevation was one of the strongest static predictors for water temperature.

This is physically reasonable because elevation is related to regional temperature, atmospheric conditions, snowmelt, and watershed position.

3.2 Upstream Drainage Area

MERIT Hydro provides upstream drainage area.

Because upstream area is strongly right-skewed, the model uses

$$\boxed{A_i = \log\left(1 + \mathrm{UPA}_i\right).}$$

This transformation reduces the influence of extremely large drainage areas while preserving relative differences among river locations.

3.3 River Width

River width was similarly transformed as

$$\boxed{W_i = \log\left(1 + \mathrm{Width}_i\right).}$$

River width did not improve water-temperature prediction when used as a direct regression predictor and slightly reduced accuracy in the temperature ablation experiment.

However, width remained useful as part of the **graph edge weighting**.

3.4 Height Above Nearest Drainage

HAND represents height above the nearest drainage surface.

It was included as

$$H_i = \mathrm{HAND}_i.$$

For the current water-temperature experiment, HAND produced essentially no change in predictive performance and was removed from the reduced static model.

3.5 Local River Slope

Local slope was calculated from a node and its immediate downstream successor.

If i flows to j,

$$\boxed{S_i = \max\left(\frac{z_i - z_j}{d_{ij}}, 0\right),}$$

where

$$z_i - z_j$$

is the elevation difference and d_{ij} is river distance.

Local slope was one of the strongest static predictors in several experiments.

3.6 Static Predictor Ablation

For water temperature, the original seasonal graph model produced

$$R^2 = 0.7657.$$

Adding individual MERIT predictors produced the following approximate graph results:

Feature Set	**Graph R^2**	ΔR^2
Seasonal baseline	0.7657	0.0000
+Elevation	0.8364	+0.0706
+ log(UPA)	0.7760	+0.0103
+ log(Width)	0.7492	−0.0166
+HAND	0.7657	0.0000
+Local slope	0.8083	+0.0426
+All static	0.8611	+0.0953

Elevation and local slope therefore provided the largest individual improvements.

A reduced static model retained

$$\boxed{\text{Elevation}, \quad \log(\text{UPA}), \quad \text{Local slope}.}$$

For water temperature, this produced

$$\boxed{R^2_{\text{linear}} = 0.8637}$$

and

$$\boxed{R^2_{\text{graph}} = 0.8628.}$$

Thus, once the most informative physical river characteristics were explicitly included, the linear and graph models became nearly identical in predictive accuracy.

This suggests that part of the graph effect had previously been capturing unmeasured physical differences among river locations.

4 General Regression and Graph Model

The complete model family can be written as

$$
\begin{aligned}
y_{i,t} = \beta_0 &+ \sum_{p=1}^{P} \beta_p x_{i,t,p} + \sum_{q=1}^{Q} \gamma_q s_{i,q} \\
&+ \sum_{k=1}^{K_{\text{day}}} \left[a_k \sin\left(\frac{2\pi k h_t}{24}\right) + b_k \cos\left(\frac{2\pi k h_t}{24}\right) \right] \\
&+ \sum_{\ell=1}^{K_{\text{year}}} \left[c_\ell \sin\left(\frac{2\pi \ell d_t}{365.25}\right) + e_\ell \cos\left(\frac{2\pi \ell d_t}{365.25}\right) \right] \\
&+ I_{i,t} + g_i + \varepsilon_{i,t}.
\end{aligned}
$$

Here,

$$x_{i,t,p}$$

represents dynamic chemistry predictors,

$$s_{i,q}$$

represents static MERIT Hydro characteristics,

$$I_{i,t}$$

represents optional daily–annual interaction terms, and

$$g_i$$

represents the graph spatial effect.

Not every target uses every term.

Instead, this equation defines the full model family from which target-specific models are selected.

5 Graph-Laplacian Component

The river network is represented as a weighted graph.

For two connected nodes i and j, the basic distance conductance is

$$w_{ij} = \frac{1}{d_{ij}},$$

where d_{ij} is river-path distance.

The weighted adjacency matrix is

$$W = (w_{ij}),$$

and the degree matrix is

$$\boxed{D_{ii} = \sum_j w_{ij}.}$$

The Graph Laplacian is

$$\boxed{L = D - W.}$$

5.1 Graph Smoothness Penalty

The graph spatial effect is regularized by

$$\boxed{g^\top L g.}$$

Equivalently,

$$\boxed{g^\top L g = \frac{1}{2} \sum_{i,j} w_{ij} \left(g_i - g_j\right)^2 .}$$

Thus, strongly connected river nodes are encouraged to have similar graph effects.

5.2 Hydrologically Weighted Graphs

The graph edge weights can also incorporate physical river characteristics.

The general form is

$$\boxed{w_{ij} = \frac{\exp(s_{ij})}{d_{ij}}.}$$

For example, the area–width weighting has the form

$$\boxed{w_{ij} = \frac{\exp\left(\alpha_A z_{A,ij} + \alpha_W z_{W,ij}\right)}{d_{ij}},}$$

where

$$z_{A,ij}$$

represents standardized upstream-area information and

$$z_{W,ij}$$

represents standardized river-width information.

The current candidate graph schemes include

distance-only, area, width, slope, area-width.

6 Model Objective

The Graph-Laplacian model minimizes an objective of the form

$$\boxed{\sum_i \frac{1}{n_i} \sum_t \left(y_{i,t} - X_{i,t}\beta - S_i\gamma - g_i\right)^2 + \lambda_L g^\top L_{\text{eff}} g + \lambda_\beta \|\theta_{\text{pen}}\|_2^2.}$$

The factor

$$\frac{1}{n_i}$$

gives each monitoring location approximately equal total influence regardless of how many hourly measurements are available at that station.

The parameter

$$\lambda_L$$

controls graph smoothing.

A large value gives stronger smoothing across the network:

$$\lambda_L \uparrow \implies g_i \approx g_j$$

for strongly connected nodes.

The ridge parameter

$$\lambda_\beta$$

regularizes regression coefficients.

7 Kron Reduction

The full river network contains many interior nodes with no direct water-quality measurements.

The Laplacian is partitioned as

$$L = \begin{bmatrix} L_{BB} & L_{BI} \\ L_{IB} & L_{II} \end{bmatrix},$$

where

$$B$$

contains retained monitoring nodes and

$$I$$

contains interior river nodes.

The effective Laplacian is obtained using the Schur complement:

$$\boxed{L_{\text{eff}} = L_{BB} - L_{BI} L_{II}^{-1} L_{IB}.}$$

This eliminates unobserved interior nodes while preserving their effective influence on the monitoring locations.

8 Graph Compression

For Colorado, four graph resolutions were tested:

$$\boxed{67{,}401 \to 6{,}771 \to 710 \to 67}$$

nodes corresponding to

$$\text{Full}, \qquad \text{10-pixel}, \qquad \text{100-pixel}, \qquad \text{Structural}.$$

After Kron reduction to the same monitoring nodes, the effective Laplacians were numerically equivalent.

Graph	Max. Absolute Difference	Relative Frobenius Difference
10-pixel	2.029×10^{-13}	7.145×10^{-13}
100-pixel	2.063×10^{-13}	7.178×10^{-13}
Structural	2.061×10^{-13}	7.176×10^{-13}

Therefore,

$$\boxed{L_{\text{eff}}^{(\text{Full})} \approx L_{\text{eff}}^{(10)} \approx L_{\text{eff}}^{(100)} \approx L_{\text{eff}}^{(\text{Structural})}.}$$

The graph can therefore be compressed dramatically without changing the effective distance-based Laplacian at the retained monitoring locations.

9 Leave-One-Node-Out Validation

Model accuracy is evaluated using nested leave-one-node-out cross-validation.

If a target has N usable monitoring nodes, the outer loop contains

$$\boxed{N \text{ folds.}}$$

In each fold, one monitoring node is completely held out.

All observations from that location are excluded from training.

The remaining nodes are used for model fitting and hyperparameter selection.

The outer held-out node is then predicted.

The final reported

$$R^2$$

and RMSE are calculated only from these outer held-out predictions.

9.1 Inner LOO Tuning

The inner LOO procedure selects quantities such as

$$K, \qquad K_{\text{day}}, \qquad K_{\text{year}}, \qquad \lambda_L, \qquad \lambda_\beta,$$

and graph weighting scheme.

Thus,

$$\boxed{\text{outer LOO} = \text{evaluation}}$$

while

$$\boxed{\text{inner LOO} = \text{model and hyperparameter selection.}}$$

The outer held-out node is never used for tuning.

10 Current Target-Specific Models

10.1 Water Temperature

The strongest current water-temperature specification uses

$$\boxed{\text{Specific conductance} + \text{Elevation} + \log(\text{UPA}) + \text{Local slope}}$$

together with approximately

$$\boxed{K_{\text{day}} = 4, \qquad K_{\text{year}} = 4,}$$

and daily–annual interaction terms.

The current outer-LOO results are

$$\boxed{R^2_{\text{linear}} = 0.8637,}$$

$$\boxed{R^2_{\text{graph}} = 0.8628.}$$

The linear and graph models are therefore nearly tied after explicit physical river characteristics are included.

10.2 Dissolved Oxygen

The current DO model includes

$$\boxed{\text{Water temperature} + \text{Specific conductance} + \text{Elevation} + \log(\text{UPA}) + \text{Local slope.}}$$

Higher-order seasonal tuning strongly favored

$$\boxed{K = 2.}$$

For the graph model, $K = 2$ was selected in 13 of 17 outer folds.

The resulting performance was

$$\boxed{R^2_{\text{linear}} = 0.7234,}$$

$$\boxed{R^2_{\text{graph}} = 0.7261.}$$

The graph model selected distance-only weighting in 16 of 17 outer folds.

10.3 Specific Conductance

The current conductance model uses

$$\boxed{\text{Water temperature} + \text{Local slope.}}$$

Daily seasonality was found to be simple:

$$\boxed{K_{\text{day}} = 1}$$

in 23 of 24 graph folds.

Annual seasonality required greater flexibility.

The inner LOO selections were

$$K_{\text{year}} = 3 \quad \text{in } 6/24 \text{ folds},$$

$$K_{\text{year}} = 4 \quad \text{in } 9/24 \text{ folds},$$

and

$$K_{\text{year}} = 5 \quad \text{in } 9/24 \text{ folds}.$$

The graph selected area–width weighting in

$$\boxed{24/24}$$

outer folds.

The current results are

$$\boxed{R^2_{\text{linear}} = -0.1200,}$$

$$\boxed{R^2_{\text{graph}} = 0.2525.}$$

The graph reduced RMSE by approximately

$$\boxed{18.30\%}$$

relative to the corresponding linear model.

10.4 pH

The current pH model uses specific conductance and seasonal terms.

The seasonal graph model produced approximately

$$\boxed{R^2_{\text{graph}} = 0.130.}$$

Elevation was the best individual static MERIT predictor tested, but it did not materially improve the graph result beyond the seasonal graph baseline.

Three response representations were also tested:

$$\text{raw pH}, \qquad \text{standardized pH}, \qquad 10^{7-\text{pH}}.$$

Raw pH and standardized pH produced identical predictive results:

$$\boxed{R^2_{\text{graph}} = 0.1316.}$$

Modeling hydrogen-ion activity instead reduced graph accuracy to

$$R^2 = 0.0954.$$

Therefore, the conventional raw pH scale was retained.

10.5 Turbidity

Turbidity remains the least predictable target with the currently available variables.

The strongest tested linear model produced approximately

$$\boxed{R^2_{\text{linear}} = 0.0554.}$$

The best tested static graph result was approximately

$$R^2_{\text{graph}} = 0.0145.$$

Local slope was the strongest individual MERIT variable tested.

These results suggest that turbidity variation is likely driven by dynamic event-based variables that are not yet present in the model, especially rainfall and runoff.

11 Current Best Pre-Weather Results

Target	**Best Model**	R^2	**Main Structure**
Water Temperature	Linear	0.8637	$K \approx 4 +$ MERIT
Dissolved Oxygen	Graph	0.7261	$K = 2 +$ MERIT
Specific Conductance	Graph	0.2525	$K_d = 1$, flexible K_y
pH	Graph	0.1316	seasonal + graph
Turbidity	Linear	0.0554	seasonal + slope

12 Interpretation

The experiments show that temporal and spatial information play different roles across water-quality variables.

Water temperature is dominated by predictable seasonal behavior and physical geography.

Elevation and local slope provide particularly strong information, and once these variables are included explicitly, the Graph Laplacian adds little additional predictive accuracy.

Dissolved oxygen benefits from both physical covariates and a modest graph effect.

Specific conductance displays the strongest evidence for a meaningful Graph-Laplacian contribution. Its linear model remains poor while the area–width graph substantially improves unseen-node prediction.

pH shows a smaller but persistent graph effect.

Turbidity remains poorly explained by chemistry, seasonality, and static river characteristics alone.

This suggests the general interpretation

$$\boxed{\text{seasonality explains when,}}$$

$$\boxed{\text{MERIT characteristics explain physical site differences,}}$$

and

$$\boxed{\text{the Graph Laplacian captures remaining river-network spatial structure.}}$$

13 Next Stage: Weather and Hydrologic Forcing

The next stage will incorporate dynamic environmental predictors, including variables such as

$$\text{air temperature,} \qquad \text{precipitation,} \qquad \text{rainfall accumulation,}$$

and potentially

$$\text{air quality.}$$

Useful rainfall features may include

$$\text{Rain}_{1h}, \qquad \text{Rain}_{6h}, \qquad \text{Rain}_{24h}, \qquad \text{Rain}_{72h}.$$

These variables are expected to be particularly important for turbidity and specific conductance and may also improve dissolved oxygen and water temperature prediction.